Hybrid CA for Avascular Tumour Growth: From Eco–Evolutionary Selection (Gerlee & Anderson, 2007) to Nutrient-Gradient-Driven Invasion (Messina et al., 2023)

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Introduction

Why these two papers

Both papers study **avascular tumour growth** with the same modelling backbone:

- Cells: as a **cellular automaton (CA)** on a lattice;
- Nutrients: as **continuous fields**, which *drive* cell decisions (by evolution);
- Evolution: via **reaction–diffusion** (with Partial Differential Equations);
- Bidirectional coupling: cells *consume/produce* nutrients.

They differ in the biological mechanism emphasized:

- **Gerlee & Anderson (2007):** oxygen limitation drives **eco–evolutionary selection** (heritable variation in cell response).
- **Messina et al. (2023):** glucose gradients drive **directional invasion/chemotaxis** and separate drivers of **growth vs invasiveness**.

Take-home: same hybrid CA/PDE scaffold $\Rightarrow$ clean comparison of *selection under oxigen scarcity* vs *migration under gradients*.

Common mathematical scaffold (hybrid CA + PDE)

Lattice. $\Lambda = \{x_i\}_{i=1}^{N} \subset \Omega \subset \mathbb{R}^d$ (typically $d = 2$). One cell per site.

CA state. For each site x_i,

$$s_i(t) \in \mathcal{S} \quad \text{(e.g., proliferative/quiescent/apoptotic/motile).}$$

Local update (deterministic or stochastic):

$$s_i(t + \Delta t) = \mathcal{U}\big(s_i(t), \{s_j(t)\}_{j \in \mathcal{N}(i)}, \boldsymbol{c}(x_i, t); \boldsymbol{\theta}\big) .$$

Microenvironment fields. $\boldsymbol{c}(x, t) \in \mathbb{R}^m$ (oxygen, glucose, etc.) evolve by

$$\partial_t \boldsymbol{c} = \boldsymbol{D}\Delta \boldsymbol{c} + \boldsymbol{R}(\boldsymbol{c}, \{s_i\}) ,$$

where $\boldsymbol{R}$ encodes cell-dependent consumption/production.

Coupling. s_i depends on $\boldsymbol{c}(x_i, t)$; $\boldsymbol{R}$ depends on occupancy/state at sites.

Gerlee & Anderson (2007): oxygen-driven selection in a hybrid CA

Aim. Explain how oxygen availability shapes (i) tumour **morphology** and (ii) **clonal/phenotypic evolution** in avascular growth.

Key modelling choice: cell response network. Each cell carries a feed-forward map from local microenvironment to action:

$$\boldsymbol{y}_i(t) = F(\boldsymbol{u}_i(t); \boldsymbol{w}_i)\,, \qquad \boldsymbol{u}_i(t) = \boldsymbol{u}(\boldsymbol{c}(x_i, t))\,,$$

where $\boldsymbol{w}_i$ are cell-specific parameters (weights/thresholds).

Heritability + mutation. Upon division:

$$\boldsymbol{w}_{\text{child}} = \boldsymbol{w}_{\text{parent}} + \boldsymbol{\eta} \quad \text{(small random perturbation).}$$

Resource competition $\Rightarrow$ selection over $\boldsymbol{w}$ (eco–evolutionary dynamics).

Reported outcomes (qualitative). Oxygen levels shift tumour **shape** (compact vs fingered) and modulate emergent **aggressiveness/diversity**.

Messina et al. (2023): glucose gradients, chemotaxis, growth vs invasiveness

Aim. Quantify how **glucose gradients** affect spheroid **growth** and **invasiveness**, and identify dominant parameters for each.

Microenvironment: glucose field. A representative reaction–diffusion form:

$$\partial_t g = D_g \Delta g - \rho\big(s(x,t)\big)\, g,$$

with boundary conditions inducing either:

- **isotropic supply** (uniform boundary g),
- **anisotropic supply** (different boundary values) $\Rightarrow$ **gradient**.

CA dynamics with chemotaxis. Updates depend on local g and (for movement) on ∇g:

$$s_i(t+\Delta t) = \mathcal{U}\Big(s_i(t), \{s_j\}_{j\in\mathcal{N}(i)}, g(x_i,t), \nabla g(x_i,t); \boldsymbol{\theta}\Big)\,.$$

Reported outcomes (qualitative). Growth is mainly controlled by proliferation/adhesion-type parameters, while invasiveness is primarily driven by chemotactic sensitivity.

Comparison lens (what the joint story will emphasize)

Same backbone: CA on lattice + nutrient PDEs + coupling.

Different "cell policy" mechanisms:

- (2007) policy is **adaptive/evolvable**: $F(\cdot; \boldsymbol{w})$ with $\boldsymbol{w}$ mutating under selection.
- (2023) policy is **rule-based with gradient sensing**: explicit chemotaxis via ∇g.

Microenvironment as driver:

scarcity (oxygen) $\Rightarrow$ selection over strategies vs gradient (glucose) $\Rightarrow$ directed invasion.

Next (later slides): model specifics (states, boundary conditions), metrics (morphology/diversity/invasiveness), and a like-for-like comparison table.

Gerlee & Anderson (2007)

Model state and spatial discretization

Lattice (CA layer). Tissue is a 2D grid $\Lambda = \{(i,j)\}_{i,j=0}^{N-1}$ with spacing d (one cell per site).
Each site stores an automaton element:

$$S_{i,j}(t) \in \{\emptyset,\ \text{alive cell},\ \text{necrotic cell}\}.$$

Cell attributes (per occupied site). Each living cell carries (minimally):

- age $a(t)$ (for proliferation check),
- phenotype/decision mechanism: network parameters $\boldsymbol{w}$ (weights/thresholds),
- (optional) flags: quiescent/motile/metabolic mode (in full model).

Microenvironment fields (PDE layer on same grid). Continuous concentrations sampled on the same lattice:

$$c_{i,j}(t)\ (\text{oxygen}), \qquad g_{i,j}(t)\ (\text{glucose}), \qquad h_{i,j}(t)\ (\mathrm{H}^+),$$

with constant Dirichlet boundary conditions representing surrounding vasculature.

Reaction–diffusion PDEs and coupling to CA

Field dynamics (reaction–diffusion). For $i \in \{c, g, h\}$:

$$\partial_t i(\boldsymbol{x}, t) = D_i \, \Delta i(\boldsymbol{x}, t) \; \pm \; f_i(\boldsymbol{x}, t).$$

Consumption/production is cell-resolved:

$$f_i(\boldsymbol{x}, t) = \begin{cases} 0, & \text{if site empty,} \\ r_i \, F(\boldsymbol{x}), & \text{if occupied,} \end{cases}$$

where r_i are base rates and $F(\boldsymbol{x})$ is a modulated metabolic factor.

Discretization. PDEs are discretized by standard 5-point finite differences on spacing d and time step Δt:

$$\Delta i \approx \frac{i_{i+1,j} + i_{i-1,j} + i_{i,j+1} + i_{i,j-1} - 4 i_{i,j}}{d^2}.$$

Interpretation. Nutrients diffuse on the grid; each occupied lattice point contributes a localized sink/source term (via f_i).

Cell decision rule: response network + mutation

Cell "policy" as a feed-forward response network. At site (i, j) sample local inputs (e.g., neighbour count and local nutrients):

$$\boldsymbol{x}_{i,j}(t) = \boldsymbol{x}\big(S_{\mathcal{N}(i,j)}(t),\ c_{i,j}(t), g_{i,j}(t), h_{i,j}(t)\big).$$

Compute network response (abstractly):

$$\boldsymbol{y}_{i,j}(t) = F(\boldsymbol{x}_{i,j}(t);\ \boldsymbol{w}_{i,j}),$$

whose argmax (or thresholded outputs) selects action

$$A \in \{\text{Proliferate (P)},\ \text{Quiesce (Q)},\ \text{Apoptose (A)}\} \quad \text{(and optionally Move/Metabolism in full model).}$$

Heritability and mutation at division. When a cell divides, daughter inherits network parameters with perturbation:

$$\boldsymbol{w}_{\text{child}} = \boldsymbol{w}_{\text{parent}} + \boldsymbol{\eta}, \qquad \boldsymbol{\eta} \sim \mathcal{N}(\boldsymbol{0}, \sigma^2 I) \text{ on mutated entries.}$$

Simulation loop and data-structure choices

Core loop (per global time step).

1. Solve PDEs on the $N \times N$ grid (finite differences; Dirichlet boundaries).
2. Create a list of living cells and **update in random order**.
3. For each cell:
   - sample $\boldsymbol{x}$ from local environment; compute action A via network;
   - execute:
     - **P:** if age $\geq A_p$ and empty neighbour exists, place daughter in a neighbouring site; else no division;
     - **Q:** mark quiescent state;
     - **A:** remove cell (site becomes empty).
   - **Necrosis:** starvation death leaves an immobile necrotic occupant (site remains blocked).
   - (optional) **Movement:** if enabled and not divided, attempt random move to empty neighbour.

Implementation view (minimal). Two coupled arrays:

$$\texttt{gridState}[i,j] \ \ \text{(empty/alive/necrotic)} \qquad \texttt{C}[i,j] \ \ \text{(oxygen field)}$$

(+ `G`, `H` in full model), and a per-cell struct storing $(a, \boldsymbol{w}, \text{flags})$.

Results: morphology vs oxygen (hypoxia)

Control parameter. Tissue background oxygen concentration c_0 (or equivalently base oxygen consumption).

Emergent morphologies (qualitative).

- **Low oxygen (hypoxic), e.g.** $c_0 = 10$**:** diffusion-limited growth $\Rightarrow$ **fingered / branched** tumour boundary (DLA-like).
- **Higher oxygen, e.g.** $c_0 = 2.5$ **and above:** growth becomes **more regular / round**, with weaker fingering.

Mechanistic interpretation in the model. Local oxygen drop across the boundary makes proliferation a "race" towards higher c:

$$\text{division at boundary} \;\Rightarrow\; \text{local sink} \;\Rightarrow\; \text{neighbour disadvantage} \;\Rightarrow\; \text{tip growth.}$$

Onset of cell death.

- Hypoxic case: cell death appears early ($t \approx 10$, $\sim$7 days).
- Higher oxygen: hypoxic core appears later ($t \approx 35$, $\sim$23 days).

Results: growth dynamics and viable rim structure

Tumour composition over time. After cell death begins, the tumour organizes into:

- necrotic/dead core (resource-deprived),
- **thin rim** of viable cells near boundary where $c(\boldsymbol{x}, t)$ remains sufficient.

Growth curves (qualitative).

- Hypoxic case ($c_0 = 10$): after early death, **approximately linear** increase in total cell count; proliferating/quiescent counts become roughly steady (limited to boundary tips).
- Higher oxygen: tumour grows larger before core death; then continues with dead-core + viable-rim dynamics.

Interpretation: oxygen availability sets an effective "active perimeter" that limits net growth once diffusion becomes limiting.

Results: population diversity (Shannon index)

Metric. Population diversity tracked via Shannon index

$$H(t) = -\sum_k p_k(t)\,\log p_k(t),$$

with p_k the fraction of clone k.

Time evolution differs by oxygen level.

- **Low oxygen ($c_0 = 10$):** H rises rapidly to ≈ 0.9 around $t \approx 30$ ($\sim$20 days), then slowly decreases to ≈ 0.8 by $t \approx 65$ days.
- **Higher oxygen:** H increases more slowly; stays low (≈ 0.2) while a quiescent core persists ($t \lesssim 40$), then increases after death onset and approaches ≈ 0.8 by the end.

Model explanation. Fingered morphology $\Rightarrow$ spatially separated "tips" $\Rightarrow$ quasi-independent evolution $\Rightarrow$ higher early diversity; round growth $\Rightarrow$ stronger competition $\Rightarrow$ slower diversity increase.

Results: evolved phenotypes under hypoxia (aggressiveness)

Phenotype proxy. Average response probabilities of the evolved networks (Table-level summary):

$$\bar{\boldsymbol{r}} = (\bar{r}_P, \bar{r}_Q, \bar{r}_A), \quad \text{plus consumption multiplier } \bar{\gamma}.$$

Reported shift (high vs low oxygen).

- Under **normoxic/high** c_0 conditions: dominant clones remain close to ancestral response (small changes).
- Under **hypoxia/low** c_0 (e.g. $c_0 = 10$): clones evolve toward

$$\bar{r}_P \uparrow, \qquad \bar{r}_Q \downarrow, \qquad \bar{r}_A \downarrow,$$

  plus increased consumption / shorter effective cell cycle (more "aggressive" strategy).

Interpretation: harsher microenvironment $\Rightarrow$ stronger selection pressure $\Rightarrow$ survival of fast-cycling, low-apoptosis phenotypes.

Messina et al. (2023)

Domain, layers, and state variables

Two-layer lattice. Square domain $2\,\mathrm{mm} \times 2\,\mathrm{mm}$ discretized into 100×100 lattice points (LPs), $\Delta = 20\,\mu\mathrm{m}$.

Cellular layer (CA). Each LP stores occupancy and phenotype:

$$S_{x,y}(t) \in \{\emptyset,\ \text{active},\ \text{starved},\ \text{necrotic}\},$$

plus an **adhesion state** (attached vs detached) for motility rules.

Glucose layer (PDE field). Glucose concentration $C(x,t)$ sampled on each LP:

$$C_{x,y}(t) \equiv C(x_{x,y}, t).$$

Initial/boundary conditions.

- isotropic: $C(\cdot, 0) = 5.5$ mM and boundary fixed at 5.5 mM;
- gradient: boundary fixed at 8 mM (left) and 3 mM (right), with initial linear profile (mean 5.5 mM).

Initial spheroid: radius $\approx 50\,\mu\mathrm{m}$ (21 cells) at domain center; simulated time ≈ 48 h.

Glucose PDE, numerical method, and uptake term

Reaction–diffusion.

$$\partial_t C = D\,\Delta C - k(x,t),$$

with disparate time scales (diffusion seconds vs division hours) $\Rightarrow$ **adiabatic approximation**:

$$0 = D\,\Delta C - k(x,t) \qquad \text{(pseudo-stationary)}.$$

Finite differences on LP grid. Discretize Laplacian with 5-point stencil on spacing Δ.

Solver. Pseudo-stationary field solved each CA update using **SOR (simultaneous over-relaxation)** with **Chebyshev acceleration**.

Consumption (Michaelis–Menten, cell-localized).

$$k(x,t) = I(x,t)\,\frac{k_{\max}\,C(x,t)}{C(x,t)+K_m},$$

where $I(x,t) \in \{0,1\}$ indicates LPs containing an *active* or *starved* cell (implemented via indicator/Dirac-delta representation on the grid).

CA rules: phenotype switching, migration, proliferation, death

Phenotype thresholds. Two glucose thresholds $C_{\text{starv}} > C_{\text{nec}}$:

$$\begin{cases} \text{active} & C_{x,y} > C_{\text{starv}}, \\ \text{starved} & C_{\text{nec}} < C_{x,y} < C_{\text{starv}}, \\ \text{necrotic} & C_{x,y} < C_{\text{nec}} \quad \text{(irreversible)}. \end{cases}$$

Event clocks.

- migration every T_m,
- duplication every T_d,
- apoptosis if starved longer than T_{apop} (LP becomes empty).

Migration (8-neighbourhood + biased random walk). A cell is either *attached* or *detached*. For an attached cell, the probability to detach is

$$P_{\text{adh}} = \frac{N_{\text{cell}}^{\text{att}}}{K + N_{\text{cell}}^{\text{att}}}, \qquad P_{\text{adh}} = 0 \text{ if detached},$$

where $N_{\text{cell}}^{\text{att}}$ is the number of attached neighbouring cells and K controls adhesion strength.

Migration direction choice and proliferation rules

Direction choice considers the 9 candidate positions (8 neighbours + stay). Let C_5 be glucose at the current site and C_i at candidate position i. Define the relative change

$$A_i = \frac{C_i - C_5}{C_5}, \qquad S_i = e^{\alpha A_i}, \qquad s_i = \frac{S_i}{\sum_{j=1}^{9} S_j},$$

with chemotactic sensitivity α. The move direction is sampled according to $\{s_i\}_{i=1}^{9}$ (via partition of $[0, 1]$ into intervals).

Proliferation. At each T_d, an *active attached* cell places a daughter in a random empty neighbour; if none exists, a **pushing** rule shifts cells along a path toward the cluster edge to create space.

Simulation process and concrete data structures

Per-step workflow.

1. Given current cellular layer, compute consumption mask $I(x, t)$.
2. Solve pseudo-stationary glucose field $C_{x,y}$ (SOR + Chebyshev).
3. Update each cell using local $C_{x,y}$:
   - phenotype transitions (active/starved/necrotic),
   - migration attempts at multiples of T_m (adhesion + chemotaxis),
   - proliferation attempts at multiples of T_d (placement/pushing),
   - apoptosis if starved duration $> T_{\text{apop}}$.

Minimal implementation view.

- `cellState[100][100]`: enum {empty, active, starved, necrotic}
- `attachFlag[100][100]`: bool (attached/detached)
- `timerM[100][100]`, `timerD[100][100]`, `starvTime[100][100]`
- `C[100][100]`: glucose field (float), updated by SOR iterations
- Random-number stream for stochastic events (migration direction, adhesion, etc.)

Replicates (stochasticity). Same parameter set is simulated multiple times and averaged (paper uses $n = 40$ replicates after convergence analysis).

Results: isotropic vs gradient glucose fields

Two experimental in silico settings.

- **Isotropic glucose:** uniform boundary supply $\Rightarrow$ no directional cue.
- **Gradient glucose:** boundary values impose $\nabla C \neq 0$ across domain $\Rightarrow$ directional cue for chemotaxis.

Observed qualitative behaviours.

- In isotropic case: outward migration is not directionally biased; invasion patterns remain approximately symmetric.
- In gradient case: cells adapt migration and exhibit **anisotropic invasion** aligned with the glucose gradient (chemotaxis-driven).

Key message: external gradient is not required to grow, but it is a strong driver for *directed* invasion.

Results: measured outputs (growth vs invasiveness)

Growth observables. The paper tracks tumour growth via:

- total number of cells $N(t)$,
- number of cells adhered to the core $N_a(t)$ (core size/area proxy).

Invasiveness observable. Let $N_d(t)$ be the number of detached/migrated cells. Define an invasiveness ratio (as in the paper):

$$I(t) = \frac{N_d(t)}{N_a(t)}.$$

Qualitative interpretation.

- Large N_a with small N_d: compact growth (low invasion).
- Moderate N_a with large N_d: dissolving core / strong invasion.

This separation (core vs detached) is used throughout the sensitivity analysis to disentangle growth and invasion.

Results: global sensitivity analysis (parameter ranking)

Parameters varied.

$$\alpha \text{ (chemotactic sensitivity)}, \quad K \text{ (adhesion)}, \quad T_d \text{ (doubling time)}, \quad T_m \text{ (migration interval)}.$$

Global sensitivity (main findings).

- **Isotropic field:** effect of α is **negligible**.
- **Gradient field:** α becomes **fundamental** in governing invasion (direction and magnitude).
- T_d and K are **critical** for core evolution and detachment in both fields; T_m has a similar but weaker influence.

High-level conclusion.

$$\text{growth} \approx f(T_d, K) \qquad \text{whereas} \qquad \text{invasion under gradients} \approx f(\alpha).$$

Results: local sensitivity / regimes and "optimal adhesion"

Local sensitivity highlights.

- $\alpha = 0 \Rightarrow$ gradient insensitivity (no directed migration).
- Increasing $\alpha \Rightarrow$ stronger alignment with positive gradient (chemotactic response).

Role of adhesion K: regime transition. There is a threshold-like behaviour around $K \approx 0.5$:

- smaller K: migration flux can dissolve the core over time;
- larger K: adhesion promotes core integrity and supports proliferation-dominated growth.

Perturbation effects (qualitative).

- decreasing T_d (faster division) promotes rapid growth (larger N, larger N_a);
- changing T_m modulates invasion/growth rates (more frequent migration accelerates dynamics).

Reported "optimal" adhesion insight. An intermediate K yields many migrating cells while maintaining approximately constant core size (trade-off growth vs detachment).

Comparison

Model comparison (I): common backbone and state variables

Common backbone (hybrid CA/PDE).

lattice CA for cells + reaction–diffusion PDE for nutrients + bidirectional coupling

Discrete layer.

- both: 2D lattice; one cell per site; neighbourhood-based constraints.
- Gerlee–Anderson (2007): cell carries *internal parameters* $\boldsymbol{w}$ (response network).
- Messina (2023): cell carries *phenotype/attachment flags* + event timers (rule-based).

Continuous layer.

- both: nutrient field(s) on same grid; boundary conditions emulate external supply.
- GA2007: oxygen (and optionally glucose, H^+) as microenvironment drivers.
- M2023: glucose-only driver with pseudo-stationary solve per CA step.

Coupling. Cells read $c(x_i, t)$ (or $C_{x,y}$) to choose actions; occupancy generates local sinks/sources in PDEs.

Model comparison (II): decision mechanism and key control parameters

Cell decision mechanism.

- GA2007: *evolvable policy* $\boldsymbol{y} = F(\boldsymbol{x}; \boldsymbol{w})$, heritable with mutation

$$\boldsymbol{w}_{\text{child}} = \boldsymbol{w}_{\text{parent}} + \boldsymbol{\eta}.$$

- M2023: *rule-based policy* with explicit chemotaxis

$$S_i = e^{\alpha A_i}, \quad A_i = \frac{C_i - C_5}{C_5}, \quad s_i = \frac{S_i}{\sum_j S_j},$$

  and adhesion-controlled detachment

$$P_{\text{adh}} = \frac{N_{\text{cell}}^{\text{att}}}{K + N_{\text{cell}}^{\text{att}}}.$$

Dominant control knobs.

- GA2007: oxygen availability (c_0) sets selection pressure and morphology.
- M2023: α (chemotaxis), K (adhesion), T_d (doubling) separate invasion vs growth regimes.

Interpretation. Both encode microenvironment $\rightarrow$ behaviour, but via **evolution** (GA2007) vs **explicit gradient sensing** (M2023).

Results comparison (I): common qualitative patterns

Shared qualitative outcomes.

- **Microenvironment shapes morphology:** scarcity/gradients induce non-trivial tumour shapes (non-spherical boundaries, protrusions).
- **Spatial self-organization:** viable/active regions concentrate where nutrient is higher (typically near boundary or along gradient).
- **Separation of "core" vs "front":**
  - GA2007: viable proliferative rim/tips vs deprived core.
  - M2023: adhered core vs detached migrating population (tracked by $N_{\text{cell}}^{\text{core}}$, $N_{\text{cell}}^{\text{mig}}$).

Common modelling message. Coupled discrete–continuous dynamics is sufficient to generate **emergent macroscopic phenotypes** from local rules.

Results comparison (II): key differences in what "emerges" and what is measured

Main emergent phenomenon.

- GA2007: **eco–evolutionary selection** under hypoxia

$$H(t) = -\sum_k p_k \log p_k \quad \text{(clonal diversity)} \quad \text{and evolved response shifts.}$$

- M2023: **directed invasion** under glucose gradients

$$\phi = \frac{N_{\text{cell}}^{\text{core}}}{N_{\text{cell}}^{\text{mig}}} \quad (\downarrow\ \phi \Rightarrow \uparrow\ \text{invasiveness}),$$

  plus anisotropy counts (N_N, N_S, N_E, N_W) to quantify directional bias.

Parameter-to-outcome logic.

- GA2007: varying c_0 changes *selection pressure* $\Rightarrow$ phenotype distribution + morphology.
- M2023: varying (α, K, T_d, T_m) yields a *regime map* separating growth vs invasiveness (via sensitivity analysis).

Conclusions

Conclusions — Strengths and limitations (model-level)

Gerlee & Anderson (2007)

- **Strengths:** explicit eco–evolutionary mechanism: heritable, mutable cell policy $\boldsymbol{y} = F(\boldsymbol{x}; \boldsymbol{w})$ with $\boldsymbol{w}_{\text{child}} = \boldsymbol{w}_{\text{parent}} + \eta$; supports clonal diversity tracking (e.g., $H(t)$).
- **Limitations:** policy is flexible but less interpretable / harder to calibrate; directed gradient-sensing (chemotaxis) is not the primary driver (needs explicit addition for invasion-in-gradient questions).

Messina et al. (2023)

- **Strengths:** interpretable rule-based CA with explicit chemotaxis in glucose gradients and operational invasion metrics (core vs migrating populations, ϕ); systematic sensitivity analysis links parameters to outcomes.
- **Limitations:** no endogenous clonal evolution (rules/parameters do not mutate); heterogeneity is mainly parametric, not evolutionary.

Proposed unification (I): unified hybrid CA/PDE with evolvable policies

Goal: retain the common hybrid CA/PDE scaffold, add (i) evolvable decision rules (GA2007) and (ii) explicit gradient-driven migration (M2023).

State.

$$S_i(t) \in \{\emptyset, \text{active}, \text{starved}, \text{necrotic}\}, \quad \text{flags: attached/detached}, \quad \boldsymbol{w}_i \text{ (cell policy parameters)}.$$

Microenvironment fields (example: oxygen + glucose).

$$\partial_t c = D_c \Delta c - \rho_c(S)\, c, \qquad \partial_t g = D_g \Delta g - \rho_g(S)\, \frac{g}{g + K_m},$$

with boundary conditions setting (i) baseline oxygen level and (ii) optional glucose gradient.

Evolvable cell policy (GA2007-style).

$$\boldsymbol{x}_i(t) = \left[c_i,\ g_i,\ \nabla g_i,\ n_{\text{occ}},\ n_{\text{att}}, \dots\right], \qquad \boldsymbol{y}_i(t) = F(\boldsymbol{x}_i(t); \boldsymbol{w}_i)\,,$$

where $\boldsymbol{y}_i$ selects $A \in \{\text{Prolif}, \text{Quiesce}, \text{Die}, \text{Move}\}$.

Inheritance. On division:

$$\boldsymbol{w}_{\text{child}} = \boldsymbol{w}_{\text{parent}} + \boldsymbol{\eta}.$$

Proposed unification (II): chemotactic move operator + measurable outputs

Chemotactic move operator (M2023-style). If $A = \text{Move}$, evaluate the 9 candidate positions (8 neighbours + stay). Let g_5 be glucose at the current site and g_k at candidate k:

$$A_k = \frac{g_k - g_5}{g_5}, \qquad S_k = e^{\alpha_i A_k}, \qquad p_k = \frac{S_k}{\sum_j S_j}.$$

Sample move direction with probabilities $\{p_k\}$, subject to occupancy constraints.

Adhesion-controlled detachment (M2023-style). For attached cells:

$$P_{\text{adh}} = \frac{N_{\text{cell}}^{\text{att}}}{K + N_{\text{cell}}^{\text{att}}}.$$

Detachment state affects motility and cluster integrity.

What can be tested/quantified in the unified model.

- **Evolution:** clonal diversity $H(t) = -\sum_k p_k \log p_k$, distribution of $\boldsymbol{w}$ (and derived traits such as α_i).
- **Invasion:** core vs migrating populations and invasiveness metric

$$\phi = \frac{N_{\text{cell}}^{\text{core}}}{N_{\text{cell}}^{\text{mig}}} \quad (\downarrow \phi \Rightarrow \uparrow \text{ invasion}).$$

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